EXPERIMENT NO 3:

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Aim:

To implement and demonstrate the ID3 (Iterative Dichotomiser 3) Decision Tree Algorithm using a training dataset and classify a new sample.

Procedure:

1. **Define the training dataset in the program.**
2. **Separate the features and the target class.**
3. **Encode the categorical data into numerical values.**
4. **Train the Decision Tree model using the Entropy criterion (ID3).**
5. **Display the generated decision tree.**
6. **Predict the class label for a new sample.**
7. **Display the predicted result.**

Program:

```
# ID3 Decision Tree Algorithm
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
# Training Dataset
data = {
   'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain',
           'Rain', 'Overcast', 'Sunny', 'Sunny', 'Rain',
           'Sunny', 'Overcast', 'Overcast', 'Rain'],
   'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool',
              'Cool', 'Cool', 'Mild', 'Cool', 'Mild',
              'Mild', 'Mild', 'Hot', 'Mild'],
   'Humidity': ['High', 'High', 'High', 'High', 'Normal',
```

```
            'Normal', 'Normal', 'High', 'Normal', 'Normal',
            'Normal', 'High', 'Normal', 'High'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak',
         'Strong', 'Strong', 'Weak', 'Weak', 'Weak',
         'Strong', 'Strong', 'Weak', 'Strong'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes',
         'No', 'Yes', 'No', 'Yes', 'Yes',
         'Yes', 'Yes', 'Yes', 'No']
}
df = pd.DataFrame(data)
# Encode categorical data
le = LabelEncoder()
X = df.drop('Play', axis=1)
y = le.fit_transform(df['Play'])
encoders = {}
for col in X.columns:
    encoders[col] = LabelEncoder()
    X[col] = encoders[col].fit_transform(X[col])
# Train ID3 Decision Tree
model = DecisionTreeClassifier(criterion='entropy')
model.fit(X, y)
# Print Decision Tree
print("Decision Tree:\n")
print(export_text(model, feature_names=list(X.columns)))

# New sample for prediction
new_sample = pd.DataFrame({
    'Outlook': ['Sunny'],
    'Temperature': ['Cool'],
    'Humidity': ['High'],
    'Wind': ['Strong']
```

```
})
for col in new_sample.columns:
    new_sample[col] = encoders[col].transform(new_sample[col])
prediction = model.predict(new_sample)
print("\nPrediction:")
print("Play =", le.inverse_transform(prediction)[0])
```

Output:

```
Decision Tree:

|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook >  0.50
|   |--- Humidity <= 0.50
|   |   |--- Outlook <= 1.50
|   |   |   |--- Wind <= 0.50
|   |   |   |   |--- class: 0
|   |   |   |--- Wind >  0.50
|   |   |   |   |--- class: 1
|   |   |--- Outlook >  1.50
|   |   |   |--- class: 0
|   |--- Humidity >  0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- Temperature <= 1.00
|   |   |   |   |--- class: 0
|   |   |   |--- Temperature >  1.00
|   |   |   |   |--- class: 1
|   |   |--- Wind >  0.50
|   |   |   |--- class: 1


Prediction:
Play = No
```

Result:

The ID3 Decision Tree Algorithm was successfully implemented using the given dataset. A decision tree was generated based on Information Gain (Entropy), and the new sample was successfully classified.

Predicted Class: Play = No